Off-by-One Errors in Sequence Indexing

Directions

- Every problem shows work in which the index is off by exactly one. Repair the computation, and where asked, say what the miscount was.
- Two counts get confused all through this topic: the number of **terms** and the number of **jumps** between them. A run of terms always has one more term than jump.
- So reaching term n from term 1 takes $n-1$ jumps, and the list $a_j, \ldots, a_k$ holds $k-j+1$ terms.
- Show the corrected arithmetic beside each blank.

1. An arithmetic sequence has $a_1 = 5$ and step $d = 3$. A student wrote

$$a_7 = 5 + 7(3) = 26.$$

(a) Find a_7 correctly. $a_7 =$ _________

(b) Name the error in the student's line.

2. In an arithmetic sequence $a_4 = 20$ and $a_9 = 45$. A student wrote

$$d = \frac{45 - 20}{4} = 6.25, \qquad \text{“terms 4 to 9 — that is 4 jumps”}.$$

Find the step correctly. $d =$ _________

3. A sequence is given by the rule $a_n = 4n + 1$ for $n \geq 1$. A student reported the first term as 1, "because that is what you get at $n = 0$".

(a) Find the first term. $a_1 =$ _________

(b) Find the fifth term. $a_5 =$ _________

(c) Name the error in the student's reasoning.

4. A student needs the number of terms in the list $a_6,\ a_7,\ a_8,\ \ldots,\ a_{20}$ and writes $20 - 6 = 14$.

(a) How many terms are in the list? count = _________

(b) Explain why subtracting the two index numbers is not the count.

5. A sequence has $a_1 = 12$ and step $d = -3$. A student wrote the rule as $a_n = 12 - 3n$ and concluded that the first term is 9.

(a) Find the eighth term correctly. $a_8 =$ _________

(b) Which term equals -18? $n =$ _________

6. A sequence has $a_1 = 3$ and step $d = 4$, so $a_n = 3 + 4(n - 1)$. Asked for the sum of the first six terms, a student added the values of that rule for $n = 0, 1, 2, \ldots, 6$.

(a) Find the sum of the first six terms. sum = _________

(b) Find the total the student actually computed. total = _________

(c) Explain why the two totals differ by the amount they do.

7. An auditorium has 14 seats in row 1 and 2 more seats in each row after that. A student wrote that row 20 has $14 + 20(2) = 54$ seats.

(a) How many seats are in row 20? seats = _________

(b) Which row has exactly 46 seats? row = _________

8. A sequence is defined by $a_1 = 2$ and $a_n = a_{n-1} + 7$ for $n \geq 2$. A student claims that $a_{100} = 2 + 100(7) = 702$, and that the block of terms a_1 through a_{10} contains 9 terms.

(a) Find a_{100} correctly. $a_{100} =$ _________

(b) How many terms are in the block a_1 through a_{10}? count = _________

(c) In one sentence, say what the two mistakes have in common.